

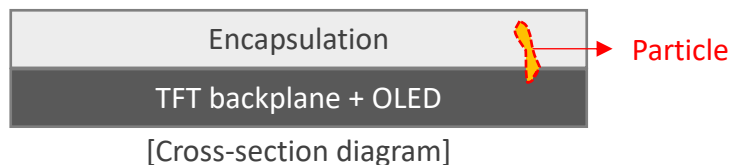
Reduced PECVD Particle Defects via RPSC NF3 Flow Optimization

Problem

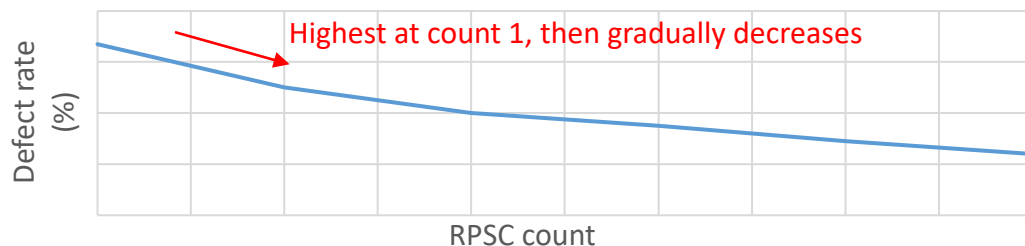
- Persistent particle defects
 - Recurring particle defects were observed
 - The defects caused reliability failures, scrap, and yield loss

Investigation

- FIB-SEM analysis
 - Identified the PECVD encapsulation layer as the defect source



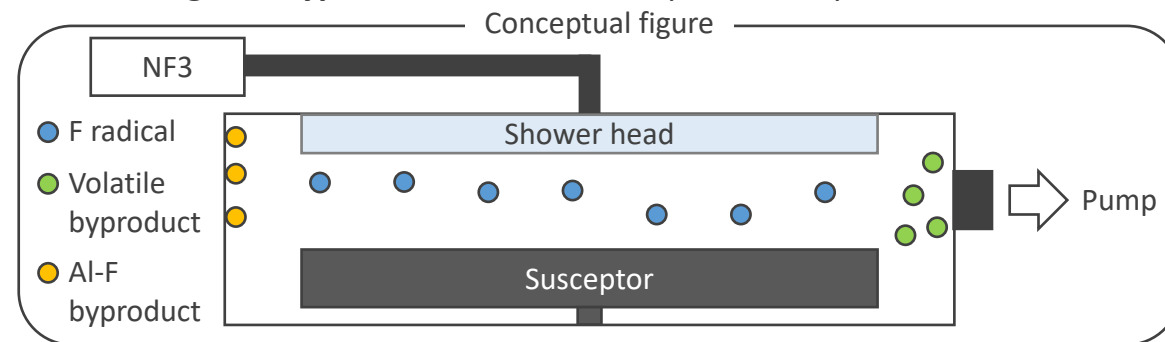
- Defect trend analysis by RPSC count
 - The defect rate was highest at RPSC count 1 and gradually decreased over subsequent counts



- This trend suggests that **post-clean residual byproducts** were depleted during the early RPSC counts
- After a fixed number of counts, chamber cleaning is performed, and the count sequence restarts

Hypothesis

- NF₃-chamber material interaction
 - NF₃ species reacted with Al-containing chamber materials during RPSC, forming **Al-F byproducts** that acted as post-clean particle sources



- Volatile Si-related byproducts (e.g., SiF₄) are pumped out, whereas Al-F byproducts can remain as **post-clean particle sources**

Verification

- Lower NF₃ flow
 - Reduced particle defects during the early RPSC counts
- EDS analysis
 - Showed strong Al and F signals, supporting the proposed chamber-material fluorination mechanism

Action

- NF₃ flow optimization
 - Adjusted the NF₃ flow during RPSC to suppress post-clean particle spikes and reduce the overall defect rate